Practical No. 11 – Two Sample Hypothesis Testing for Difference of Population Means

Blue = given data (inputs). Black = formulas. Each test uses the critical-value method, the p-value method, and (Ex.3, 4) Excel's built-in TTEST as a cross-check.

Ex.1 Light bulbs – two-sample z-test (right-tailed, large samples)

H0: μ1 = μ2 vs H1: μ1 > μ2 (company's bulbs superior; n1, n2 ≥ 30 → z-test)

Sample 1: mean x̄1	1647	
Sample 1: SD s1	27	
Sample 1: size n1	40	
Sample 2: mean x̄2	1638	
Sample 2: SD s2	31	
Sample 2: size n2	40	
Significance level α	0.05	
Standard error	6.5000	*√(s1²/n1 + s2²/n2)*
Test statistic z	**1.3846**	*z = (x̄1 − x̄2)/SE*
Critical value z(α)	1.6449	*NORMSINV(1−α)*
p-value (right tail)	0.08309	*1−Φ(z)*
Decision (critical value method)	**Fail to reject H0**	
Decision (p-value method)	**Fail to reject H0**	

Conclusion: No significant evidence that the company's bulbs last longer than the competitor's.

Ex.2 Photocopier repair times – two-sample z-test (two-tailed)

H0: μ1 = μ2 vs H1: μ1 ≠ μ2 (n1 = n2 = 75, large samples → z-test)

Sample 1: mean x̄1	83.2	
Sample 1: SD s1	19.3	
Sample 1: size n1	75	
Sample 2: mean x̄2	90.8	
Sample 2: SD s2	21.4	
Sample 2: size n2	75	
Significance level α	0.05	
Standard error	3.3276	*√(s1²/n1 + s2²/n2)*
Test statistic z	**-2.2840**	*z = (x̄1 − x̄2)/SE*
Critical value ±z(α/2)	1.9600	*NORMSINV(1−α/2)*
p-value (two-tailed)	0.02237	*2×(1−Φ(|z|))*
Decision (critical value method)	**Reject H0**	
Decision (p-value method)	**Reject H0**	

Conclusion: Reject H0 – the average repair times of the two kinds of equipment differ.

Ex.3 Two independent samples – pooled t-test (equal variances, two-tailed)

H0: μ1 = μ2 vs H1: μ1 ≠ μ2 (normal populations, equal variances, small samples → pooled t-test)

Sample I

25.00	30.00	28.00
34.00	24.00	20.00
13.00	32.00	22.00
38.00		

Sample II

40.00	34.00	22.00
20.00	31.00	40.00
30.00	23.00	36.00
17.00		

Significance level α	0.05	
n1	10	*COUNT*
n2	10	*COUNT*
Mean x̄1	26.6000	*AVERAGE*
Mean x̄2	29.3000	*AVERAGE*
Variance s1²	54.0444	*VAR (n−1)*
Variance s2²	70.0111	*VAR (n−1)*
Degrees of freedom	18	*n1 + n2 − 2*
Pooled variance sp²	62.0278	*((n1−1)s1²+(n2−1)s2²)/df*
Test statistic t	**-0.7666**	*t = (x̄1−x̄2)/√(sp²(1/n1+1/n2))*
Critical value ±t(α/2, df)	2.1009	*TINV(α, df)*
p-value (two-tailed)	0.45327	*TDIST(|t|, df, 2)*
Excel TTEST p-value (check)	0.45327	*TTEST(arr1, arr2, 2, 2) – type 2 = equal var*
Decision (critical value method)	**Fail to reject H0**	
Decision (p-value method)	**Fail to reject H0**	

Conclusion: Means of the two populations are the same at the 5% level.

Ex.4 Catalyst product volumes – Welch t-test (unequal variances, two-tailed)

H0: μ1 = μ2 vs H1: μ1 ≠ μ2 (variances unequal → Welch's t-test)

Catalyst 1 (gal)

0.63	2.64	1.85
1.68	1.09	1.67
0.73	1.04	0.68

Catalyst 2 (gal)

3.71	4.09	4.11
3.75	3.49	3.27
3.72	3.49	4.26

Significance level α	0.05	
n1	9	*COUNT*
n2	9	*COUNT*
Mean x̄1	1.3344	*AVERAGE*
Mean x̄2	3.7656	*AVERAGE*
Variance s1²	0.4548	*VAR (n−1)*
Variance s2²	0.1089	*VAR (n−1)*
s1²/n1	0.05054	
s2²/n2	0.01210	
Welch degrees of freedom	11.62	*(a+b)²/(a²/(n1−1)+b²/(n2−1))*
Test statistic t	**-9.7138**	*t = (x̄1−x̄2)/√(s1²/n1+s2²/n2)*
Critical value ±t(α/2, df)	2.1788	*TINV(α, df) with df rounded*
p-value (two-tailed)	0.0000006	*TTEST(arr1, arr2, 2, 3) – type 3 = unequal var*
Decision (critical value method)	**Reject H0**	
Decision (p-value method)	**Reject H0**	

Conclusion: Extremely strong evidence that mean product volumes differ between the two catalysts.